# Sejal Agarwal

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## PROFESSIONAL SUMMARY

Software engineer and Master's Computer Science candidate at the University of Waterloo with 2+ years of professional experience building and shipping customer-facing web and UI applications. Experienced with React, TypeScript, experimentation, and AI-powered interfaces, with a strong background in HCI and UX. Experienced collaborating across engineering, product, design, and research to turn complex workflows into intuitive user experiences.

## EDUCATION

**Master of Mathematics (Computer Science), thesis-based**
University of Waterloo — Ontario · Final CGPA 4.0/4.0 · Research in Human-Computer Interaction · September 2025 — Present

**Bachelor of Software Engineering with AI specialization**
University of Victoria — British Columbia · Final CGPA 4.0/4.0, with distinction. · September 2020 — April 2025

## SKILLS

**Frontend**: TypeScript, JavaScript, React, Next.js, Angular, HTML/CSS
**Backend & Data**: Node.js, PostgreSQL, SQL, REST APIs
**AI & Intelligent Interfaces**: RAG, LLM applications, Claude Code, GitHub Copilot, ChatGPT, Codex, Gemini
**UX & Product**: Figma, HCI, A/B testing, experimentation, user research, telemetry-driven development
**Languages**: Python, Java, Kotlin, C
**Tools**: Git, GitHub, Jira, Confluence, Microsoft Suite, Linux/UNIX, macOS, Windows

## EXPERIENCE

**Research Intern, Agentic Canvas Interactions**
Autodesk — September — December 2026 | Toronto, ON
- Design and engineer responsive, interactive canvas prototypes exploring direct-manipulation interfaces for real-time agentic workflows.
- Partner across research, design, and engineering teams to translate interaction concepts into scalable UI patterns for AI-assisted workflows.
- Conduct literature reviews on agentic workflows and direct-manipulation spatial interfaces to inform interaction design and prototype development.

**Software Engineer Intern**
Outschool — January — April 2024, September — December 2024 | San Francisco, CA
- Increased new-buyer engagement/sign-ups by **25%** by designing and implementing A/B-tested UI/UX experiments across key customer workflows.
- Built and iterated on production web features, using experimentation telemetry and user behavior to evaluate performance and guide UI improvements.
- Collaborated with product/data teams to translate user and business needs into customer-facing experiences and data-driven product recommendations.

**Software Engineer Co-op**
Open Ocean Robotics — May — August 2023 | Victoria, BC
- Improved maintainability across a large-scale web codebase by migrating 40% of the repository (~60,000 lines of code) from JavaScript to TypeScript.
- Built a responsive web image carousel for live Unmanned Surface Vehicle feeds to optimize image delivery and reduce live-streaming costs by 20%.

**Software Engineer Intern**
Respondent — September — December 2022 | New York, NY
- Refactored ~20% of an Angular web application, removing legacy front-end code and shipping 5 new user-facing features to improve platform maintainability and stability.
- Expanded Jest unit test coverage across new and existing functionality, improving frontend reliability and reducing code issues before review.

**Full-Stack Developer Co-op**
19Labs — January — April 2022 | San Mateo, CA
- Drove a platform-wide user migration for the GALE telehealth platform, consolidating 7 permission roles into 4, remapping users, and redesigning role-based UI and permissions for each user type.
- Integrated Bluetooth connectivity to stream real-time health metrics from medical devices, enabling practitioners to review live patient data.

## PROJECTS

**Otter: RAG Study Workspace + Student GenAI usage Dashboard for Instructors | 2026**
- Built a Next.js and PostgreSQL RAG workspace that provides source-grounded answers linked to exact lecture slides and page numbers, using AI development tools including Claude and Codex to accelerate implementation.
- Engineered the instructor dashboard with anonymized analytics and automated weekly query summaries, backed by a PDF vector-ingestion pipeline.

**Martletplace: UVic Community Marketplace | 2025**
- Led a 14-person engineering team as technical project lead to design and ship a full-stack marketplace web application.

## RESEARCH & PUBLICATIONS

**Thinking Inside the Box: Building UIs with Question-Guided Inputs | Graphics Interface Conference 2026**
- Designed and built an interactive question-guided UI system for creating web applications, evaluated through an HCI user study and published as a first-author paper at Graphics Interface 2026.